

MODULE 3 - ADVANCED LEVEL COMPLETE DETAILED EXPLANATIONS

1. Poisson Approximation to Binomial

Binomial distribution formula:

$$P(X=x) = {}^nC_x p^x (1-p)^{(n-x)}$$

When n is very large, p is very small and $\lambda = np$ is finite, the Binomial distribution approaches the Poisson distribution.

Approximation formula:

$$P(X=x) \approx e^{-\lambda} \lambda^x / x!$$

This approximation is mainly used for rare events occurring in large populations.

Typical rule: $n \geq 50$ and $p \leq 0.1$.

2. Central Limit Theorem (CLT)

If $X_1, X_2, \dots, X_n$ are independent and identically distributed with mean μ and variance σ^2 , then for large n , the sampling distribution of the mean is approximately Normal.

Standardization formula:

$$Z = (\bar{X} - \mu) / (\sigma / \sqrt{n}) \approx N(0,1)$$

This theorem explains why Normal distribution appears frequently in statistics, even when the original population is not Normal.

3. Gamma Distribution

Defined for $x > 0$.

PDF:

$$f(x) = (\lambda^\alpha x^{\alpha-1} e^{-\lambda x}) / \Gamma(\alpha)$$

$$\text{Mean} = \alpha / \lambda$$

$$\text{Variance} = \alpha / \lambda^2$$

If $\alpha = 1$, Gamma distribution reduces to Exponential distribution.

Gamma models waiting time for multiple independent exponential events.

4. Normal Distribution (Advanced Understanding)

Continuous, symmetric, bell-shaped distribution.

PDF:

$$f(x) = \frac{1}{\sqrt{2\pi \sigma^2}} e^{\{-(x-\mu)^2/(2\sigma^2)\}}$$

Mean = μ

Variance = σ^2

Standardization:

$$Z = (X - \mu) / \sigma$$

Normal distribution plays a central role due to CLT.

5. Variance of Binomial Distribution

If X is sum of n independent Bernoulli(p) variables:

$$X = X_1 + X_2 + \dots + X_n$$

$$\text{Mean} = np$$

$$\text{Variance} = np(1-p)$$

Independence removes covariance terms, allowing variance to add.

6. Deep Understanding of Independence

Two events A and B are independent if:

$$P(A \cap B) = P(A)P(B)$$

This implies conditional probability form:

$$P(A|B) = P(A)$$

Independence means occurrence of one event does not change the probability of the other.
Independence is different from mutually exclusive events.